

Name: _____

Class: _____

Mr. Rawson • GCSE Computer Science



Year 11 2026

Test 1: Review Topics

Test: Thursday 1 October — 35 marks, 40 minutes

Tick a box when you can do it without looking at your notes. The paper may show pseudocode or OCR's ERL — you may always answer in Python.

1. Data types

- ☐ Name all five data types by their OCR name — Integer, Real, Boolean, Character, String — and say what each one holds.
- ☐ `input()` always gives you a String, even when the user types a number.
- ☐ Cast between types: `int()` or `float()` turn text into a number, `str()` turns a number into text. `int()` cannot swallow a decimal — use `float()` when you need one.
- ☐ `"3" * 4` gives 3333, with no error at all — that is a logic error, the hardest kind to find.

2. Programs — sequence, selection and iteration

- ☐ Know the four flowchart shapes OCR uses: Terminal (start/stop), Process (do something), Input/Output (anything in or out), Decision (asks a question).
- ☐ Name the construct: the test is not *is there a condition?* — it is *does anything happen more than once?* “Keeps asking until you get it right” is Iteration, not Selection.
- ☐ Write and trace a count-controlled loop (`for`) and a condition-controlled loop (`do / until` or `while`), including one combined with an `if / elif / else`.
- ☐ Find a syntax error (breaks the rules of the language) versus a logic error (runs, but gives the wrong answer), and say which is which.

3. Manipulating strings and numbers

- ☐ Join text and values with concatenation, casting a number to a string with `str()` first.
- ☐ Every character has a position, counting from 0. Pull characters out: `length` (how many characters), `left/right` (from one end), `substring` (from the middle).
- ☐ The exam writes these in OCR's ERL, not Python — read it, don't type it: `.length` → `len()`, `.left(4)` → `[:4]`, `.right(3)` → `[-3:]`, `.substring(3,5)` → `[3:8]`.
- ☐ ERL's `substring` counts how many characters; Python's slice says where to stop. They can land in the same place — check which one a question means.
- ☐ Convert between a character and its code: `ASC('A')/ord('A') = 65`; `CHR(97)/chr(97) = a`.
- ☐ The seven operators: `+` `-` `*` `/`, then `DIV///` and `MOD/%` together answer “how many, and what is left over” (`7 DIV 2` is 3, `7 MOD 2` is 1), and `~/**` for exponent.

4. Arrays

- ☐ An array is one variable holding many values of the same type, each in a numbered box called its index, starting at 0 — the same idea as a string's numbered characters, one whole value per box instead of one character.
- ☐ Read a box with the array's name and index in square brackets; count the boxes with `len()`.
- ☐ Use a loop to visit every box, e.g. to total or search its values.

Remember: a syntax error breaks the rules of the language; a logic error runs but gives the wrong answer an index starts at 0 trace it line by line, one variable at a time

Practise with the Programming Review Packet on your class page.